Securing AI and Advanced Topics

Today's goal • 3/3 items

Module 1
Course Introduction

Module 2
Fraud Prevention with Cloud AI Solutions

Module 3
Introduction to Generative Adversarial Attacks (GANs)

Module 4
GANs and Adversarial Attacks

Understanding Black-box and White-box Adversarial Attacks

- Adversarial Attacks Explained
  Video • 8 min
- Reading References
  Reading • 10 min
- Understanding Black-box and White-box Adversarial Attacks
  Practice Assignment • Grade: 100%

Practical Implementation of Adversarial Attacks

- Hands-On Adversarial Attacks
  Video • 9 min
- Reading References
  Reading • 10 min
- Practical Implementation of Adversarial Attacks
  Practice Assignment • Grade: 100%

Module-end Assessments

- Self-Reflective Reading: GANs and Adversarial Attacks
  Reading • 45 min
- **Practice Lab: Generating Synthetic QR Codes with the Trained Generator**
  Lab • 1h
- GANs and Adversarial Attacks
  Graded Assignment • 1h

Module 5
Reinforcement Learning

Reinforcement Learning and Its Applications

- Reinforcement Learning
  Video • 8 min
- Reading References
  Reading • 10 min
- Reinforcement Learning and its Applications
  Practice Assignment • 15 min

Using RL for Adversarial Attacks and Optimizing Datasets

- Evading Malware Detectors with RL
  Video • 6 min
- Reading References
  Reading • 10 min
- Using RL for Adversarial Attacks and Optimizing Datasets
  Practice Assignment • 15 min

Module-end Assessments

- Self-Reflective Reading: Understanding Reinforcement Learning
  Reading • 45 min
- Reinforcement Learning
  Graded Assignment • 1h

Module 6
Evaluating AI Models and Performance

Feature Engineering and Model Optimization Techniques

- Challenges with Using AI for Cybersecurity
  Video • 8 min
- Reading References
  Reading • 10 min
- Feature Engineering and Model Optimization Techniques
  Practice Assignment • 15 min

Evaluating AI Models and Performance Metrics

- Evaluating AI Models
  Video • 7 min
- Reading References
  Reading • 10 min
- Evaluating AI Models and Performance Metrics
  Practice Assignment • 15 min

Module-end Assessments

- Self-Reflective Reading: Feature Engineering in Cybersecurity Applications
  Reading • 45 min
- Evaluating AI Models and Performance
  Graded Assignment • 1h

Lab

Practice Lab: Generating Synthetic QR Codes with the Trained Generator



Launch Lab | Mark as completed

Instructions

To get started, click the "**Launch Lab**" button. This will open the lab environment in a new browser tab, allowing you to interact with the virtual lab setup provided for this course.

If you prefer to work offline, you can download the lab notebook to your local environment by clicking on the file below.

Hands-on Lab - Generating Synthetic QR Codes with the Trained_Generator
IPYNB File

The lab notebook (.ipynb file) includes important instructions, questions, and coding tasks to guide you through the lab.

Be sure to follow the directions in the notebook closely to make the most of your learning experience. Enjoy exploring and experimenting in the virtual lab!

Report an issue

Go to next item →